

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

November 2013

EE 301 Electrical Machines - II

Date: 18.11.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 A three phase, 15 kW, 400 V, six pole induction motor gave the following results: [10]
No-load test: 400 V, 11.2 A, $\text{pf} = 0.19$
Short-circuit test: 100 V, 25.5 A, $\text{pf} = 0.39$
Draw the circle diagram. From this diagram, find the line current, slip, efficiency, power factor at full load. Also, find the maximum torque in terms of full-load torque. Assume that the rotor copper loss at standstill is half the total copper loss.
- Q - 2 (a) A three phase, four pole, 60 kW, 50 Hz induction motor connected to rated supply [06]
voltage and running without load consumes 3 kW. When prevented from rotating it draws rated current at 30 per cent rated supply and takes a power input of 4 kW. Assuming that under rated load conditions, the stator and rotor copper losses are equal and that the mechanical losses are 30 per cent of the no-load losses. Determine (a) slip at rated load and (b) starting torque of the motor with rated applied voltage.
- Q - 2 (b) Fill in the blanks [09]
- (i) The short circuit current at normal voltage is five times the full load current. The load slip is 4 %. The starting torque when star-delta switch is used is equal to _____ times full load torque.
 - (ii) A 4 pole, 50 Hz, 3 phase induction motor has blocked rotor reactance per phase, which is four times the rotor resistance per phase. The speed at which maximum torque develops is _____ rpm.
 - (iii) A starting torque of 100 N-m is developed by an autotransformer starter with a tapping of 40 per cent. If tapping of the autotransformer is at 80 per cent, the starting torque will be _____ N-m.
 - (iv) The power input to a 415 V, 50 Hz, six pole, three phase induction motor running at 975 rpm is 40 kW. The stator losses are 1 kW and friction and windage losses 2 kW. The efficiency of the motor is _____ per cent.
 - (v) An eighth- pole wound rotor induction motor operating on 60 Hz supply is driven at 1800 rpm by a prime mover in the opposite direction to the revolving magnetic field. The rotor current frequency is _____ Hz.
 - (vi) The frequency of rotor emf of an eighth-pole induction motor is 2 Hz. If supply frequency is 50 Hz, then the motor speed is _____ rpm.

OR

- Q - 2 (a) Draw and explain the speed torque characteristics for poly phase induction machines. [05]
- Q - 2 (b) Explain briefly the production of three phase rotating magnetic field. [05]
- Q - 2 (c) Explain the effect of time and space harmonic field in induction machines. [05]
- Q - 3 (a) Explain the construction, principle, advantages, disadvantages and application of linear induction motor. [05]
- OR**
- Q - 3 (a) State various methods used for speed control of poly phase induction machines and explain any one method. [05]
- Q - 3 (b) Explain the double cage induction motor and draw its equivalent circuit. [05]

SECTION - II

- Q - 4 Draw and explain the complete equivalent circuit of single phase induction motor based on double revolving field theory. [07]
- Q - 5 (a) Write a technical note on Schrage motor. [08]
- OR**
- Q - 5 (a) Write a technical note on Stepper motor. [08]
- Q - 5 (b) Explain the construction and working of shaded pole induction motor. [05]
- OR**
- Q - 5 (b) Write a technical note on Hysteresis motor. [05]
- Q - 6 (a) Explain the various types of induction generator [07]
- Q - 6 (b) The equivalent circuit parameters of a 230 V, 50 Hz, single phase induction motor having friction, windage loss and core loss of 50 W are given below:
 $R_{1m} = 2.4 \Omega$, $R_2' = 4.7 \Omega$, $X_{1m} = 3.2 \Omega$, $X_2' = 2.8 \Omega$ and $X_m = 90 \Omega$. Calculate (i) input current, (ii) power factor, (iii) developed power (iv) output power and (v) efficiency for a slip of 0.04. [08]
- OR**
- Q - 6 (b) The test results of a 230 V, single phase induction motor are given below: [08]
 Blocked rotor test: 110 V, 9.5 A, 450 W
 No load test: 230 V, 4.4 A, 120 W
 The starting winding is kept open during blocked rotor test and stator winding resistance is 1.4Ω . Find the equivalent circuit parameters and the core, friction and windage losses.
